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Transparency in the Use of AI Tools

AI-assisted tools were used to consolidate repository documentation, check proposal consistency, and draft this advisor handout. The author remains responsible for the final research scope, technical claims, citations, and submitted document.

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1 Current State and Thesis Claim

Next-best-view selection is often optimized through coverage, uncertainty, or information-gain proxies. These proxies help exploration, but they do not directly test whether a selected view improves the geometry of a specific target. This gap matters for egocentric indoor reconstruction: broad scene coverage can still leave a target poorly reconstructed because support is sparse, occluded, or observed from unfavorable directions.

ARIA-NBV studies a finite-candidate, target-conditioned version of this problem on Project Aria / ASE data [Eng+23, Met25] with frozen EFM3D/EVL evidence [Str+24]. The prior seminar implementation provides scene-level oracle RRI labels and a one-step scoring substrate. The thesis extends this substrate to leakage-safe target labels, fixed-budget oracle lookahead, and actor-visible finite-horizon value learning. Continuous control, Habitat/Isaac experiments, SceneScript-style global memory, VLM planning, and real-device guidance are contrast or future work unless the finite-candidate experiment first succeeds.

THESIS CLAIM

Target-specific reconstruction-quality improvement can serve as a finite-horizon NBV objective for egocentric indoor reconstruction. The thesis tests this by first measuring oracle lookahead headroom over one-step target-RRI selection. If headroom exists, an actor-visible finite-candidate value model Q_H is trained to recover it under the same candidate and acquisition budgets.

2 Formal Model and Research Questions

Let $\mathbf{P}_t$ denote the accumulated actor-visible geometry proxy at decision step t , initialized from the logged semi-dense reconstruction and updated during counterfactual rollouts. Let $\mathbf{F}_0^{\text{EVL}}$ be the frozen root egocentric evidence field, and let $\mathbf{z}_e$ be the actor-visible descriptor of the selected target. The actor-visible state is

$$\mathbf{s}_t^{\text{obs}} = (\mathbf{P}_t, \mathbf{F}_0^{\text{EVL}}, \mathbf{z}_e, \mathbf{h}_t, b_t). \quad (1)$$

The privileged oracle state augments this with GT geometry and all-candidate oracle render/evaluation assets:

$$\mathbf{s}_t^{\text{oracle}} = \left(\mathbf{s}_t^{\text{obs}}, \mathbf{M}_{\text{GT}}, \{\mathbf{M}_e^{\text{GT}}\}_{e \in \mathcal{E}}, \{\mathbf{P}_{t,i}^{\text{cand}}\}_{i=1}^{N_q} \right). \quad (2)$$

For the V1 main result, GT meshes, GT OBBs, GT crops, and all-candidate GT renders are label and evaluation assets only. V0 may use GT OBB input only as a sanity or upper-bound path, not as the main actor-visible result. The counterfactual state keeps root EVL fixed:

$$\mathbf{s}_t^{\text{cfo}} = (\mathbf{F}_0^{\text{EVL}}, \mathbf{P}_t, \mathbf{z}_e, \mathbf{h}_t, b_t, \mathbf{Q}_t, \mathbf{m}_t, \boldsymbol{\rho}_t). \quad (3)$$

The finite action set is the masked subset of candidate rows:

$$\mathcal{U}_t = \{i \in \{1, \dots, N_q\} : m_{t,i} = 1\}, \quad \mathbf{q}_t = \mathbf{q}_{t,a_t}. \quad (4)$$

Invalidity is a constraint, not weak supervision: masks apply before argmax, temperature softmax, loss targets, and bootstrap maximization. V1 uses OBS-SEL, PRED-Q, and GT-EVAL. The actor-visible descriptor is

$$\mathbf{z}_e = \varphi(\hat{\mathbf{B}}_e, \hat{\mathbf{y}}_e, \hat{p}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \mathbf{T}_e^{\text{rel}}) \quad (5)$$

It bundles observed or predicted OBB geometry, class, confidence, projected area, semidense support, EVL support, and relative pose. GT crops are selected only after protocol-level matching:

$$\mu(\hat{e}, e) = \kappa(\hat{\mathbf{y}}_{\hat{e}}, \mathbf{y}_e) \cdot \text{IoU}_{3D}(\hat{\mathbf{B}}_{\hat{e}}, \mathbf{B}_e^{\text{GT}}) \cdot \sigma(A_{\hat{e}}^{\text{proj}}, n_{\hat{e}}^{\text{semi}}, n_{\hat{e}}^{\text{EVL}}) \quad (6)$$

$$e^* = \underset{e \in \mathcal{E}}{\text{argmax}} \mu(\hat{e}, e) \quad (7)$$

$$\text{accept iff } \mu(\hat{e}, e^*) \geq \tau_\mu \quad \text{and} \quad \mu_1 - \mu_2 \geq \tau_{\text{gap}} \quad (8)$$

Targets that fail acceptance, or whose top-1/top-2 scores are ambiguous, are counted as target-invalid protocol cases rather than low target-RRI examples.

The target error is the implemented point-mesh accuracy plus mesh-to-point completeness diagnostic, not point-cloud Chamfer distance:

$$\Delta_t^e = A_t^e + C_t^e \quad (9)$$

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_t^e + \varepsilon} \quad (10)$$

$$G_t^{(H)} = \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e \quad (11)$$

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon} \quad (12)$$

$$J_{e, \log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon) \quad (13)$$

$J_e^{(H)}$ is the primary endpoint metric, $G_t^{(H)}$ is the rollout training return, and $J_{e, \log}^{(H)}$ is an ablation for scale sensitivity only.

After selecting a valid candidate, the acquired geometry is added to the current geometry:

$$\mathbf{P}_{t+1} = \mathbf{P}_t \cup \mathbf{P}_{t,i}^{\text{cand}}. \quad (14)$$

The next candidate table $\mathbf{Q}_{t+1}$ is regenerated from the updated geometry, selected-view history, and remaining budget with the same logged mixture families, while $\mathbf{F}_0^{\text{EVL}}$ remains fixed.

The three thesis questions are tests:

RQ	Advisor-facing test
RQ1	Can target-specific point-mesh RRI be measured and matched without target leakage?
RQ2	Does bounded oracle lookahead improve fixed-budget endpoint target quality over one-step selection?
RQ3	Can an actor-visible finite-candidate value model recover measurable oracle headroom from offline rollout support?

Table 1: Research questions as falsifiable tests rather than deliverable slogans.

The headroom and recovery quantities are

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}}) \quad (15)$$

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon} \quad (16)$$

3 Planned Value-Model Design

This section is subordinate to the oracle and headroom tests. The learned model must map each valid candidate row to a finite-horizon value using only actor-visible scene, target, history, and candidate features. It must be permutation-equivariant over candidate rows, apply the hard mask before action selection and training targets, and be evaluated by oracle re-scoring rather than by its own predicted values.

The recommended first architecture is a residual finite-horizon model on top of the target-conditioned one-step scorer:

$$Q_{H,\theta}(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{q}_{t,i}) = \hat{\pi}_{\psi}^e(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{q}_{t,i}) + A_{\theta}^H(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{Q}_t, \mathbf{h}_t, i). \quad (17)$$

The residual advantage is initialized near zero, so the first Q_H model behaves like the myopic scorer until rollout evidence proves that planning adds value. The scene memory uses the frozen EVL field, accumulated actor-visible geometry, target support, and directional memory:

$$\mathbf{F}_t^{\text{scene}} = \text{Conv3D}(\mathbf{F}_0^{\text{EVL}}, \mathbf{V}(\mathbf{P}_t), \mathbf{V}_{\text{dir}}(\mathbf{P}_t), \mathbf{V}_{\text{target}}(\mathbf{z}_e)). \quad (18)$$

The target token should not be a descriptor vector alone; it should read a target-local crop from the scene memory:

$$\mathbf{T}_e = \text{MLP}_{\varphi}(\text{concat}(\mathbf{z}_e, \text{ROIAlign3D}(\mathbf{F}_t^{\text{scene}}, \hat{\mathbf{B}}_e))). \quad (19)$$

Candidate orientation and accumulated visibility are separate signals. The pose branch uses the continuous R6D rotation representation [Zho+19]:

$$\mathbf{x}_{\text{pose}}(\mathbf{q}_{t,i}) = \text{concat}(\mathbf{t}_{t,i}, \mathbf{R}_{t,i}^{6D}, \mathbf{t}_{t,i} - \mathbf{t}_e, \alpha_{t,i}^e, l_{t,i}, c_{t,i}^{\text{strategy}}). \quad (20)$$

Accumulated visibility is a separate actor-visible directional memory on $\mathbb{S}^2$. The first implementation should use a moment memory:

$$\mathbf{d}_k(\mathbf{v}) = \frac{\mathbf{c}_k - \mathbf{v}}{\|\mathbf{c}_k - \mathbf{v}\|_2} \quad (21)$$

$$\mathbf{M}_{\text{dir}}(\mathbf{v}) = \sum_{k < t} w_k(\mathbf{v}) \mathbf{d}_k(\mathbf{v}) (\mathbf{d}_k(\mathbf{v}))^\top \quad (22)$$

$$\nu_{t,i}^{\text{dir}}(\mathbf{v}) = 1 - \frac{(\mathbf{d}_{t,i}(\mathbf{v}))^\top \mathbf{M}_{\text{dir}}(\mathbf{v}) \mathbf{d}_{t,i}(\mathbf{v})}{\text{tr}(\mathbf{M}_{\text{dir}}(\mathbf{v})) + \varepsilon} \quad (23)$$

Low-order spherical harmonics are a richer directional-memory ablation, not a precondition for the first value-model test. Each candidate row then receives pose, target-relative, frustum, belief-render, directional novelty, mask/reason, and history features:

$$\mathbf{p}_{t,i} = \text{concat}(\mathbf{x}_{\text{pose}}(\mathbf{q}_{t,i}), \varphi_{\text{target-rel}}(\mathbf{q}_{t,i}, \mathbf{z}_e)). \quad (24)$$

$$\mathbf{g}_{t,i} = \text{concat}(\varphi_{\text{frustum}}(\mathbf{F}_t^{\text{scene}}, \mathbf{q}_{t,i}), \varphi_{\text{belief}}(\mathbf{P}_t, \mathbf{F}_0^{\text{EVL}}, \mathbf{q}_{t,i}), \varphi_{\text{dir}}(\mathbf{M}_{\text{dir}}, \mathbf{q}_{t,i})). \quad (25)$$

$$\mathbf{x}_{t,i} = \text{concat}(\mathbf{p}_{t,i}, \mathbf{g}_{t,i}, \varphi_{\text{valid}}(m_{t,i}, \rho_{t,i}), \mathbf{H}_t). \quad (26)$$

Candidate reasoning must be permutation-equivariant; a Set Transformer candidate encoder is the first planned implementation [Lee+19]:

$$\{\mathbf{u}_{t,i}\}_{i=1}^{N_q} = E_{\text{set}}\left(\left\{\text{concat}(\mathbf{x}_{t,i}, \mathbf{T}_e, \mathbf{H}_t)\right\}_{i=1}^{N_q}, \mathbf{m}_t\right). \quad (27)$$

The head should be dueling and mask-normalized over valid actions:

$$Q_{H,\theta}(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{q}_{t,i}) = V_\theta(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{H}_t) + A_{\theta,i}^H - \frac{1}{|\mathcal{U}_t|} \sum_{j \in \mathcal{U}_t} A_{\theta,j}^H. \quad (28)$$

The thesis-safe order is therefore: target-conditioned one-step scorer, residual Q_H without a view-render branch, actor-visible belief-render branch, R6D pose features, $\mathbb{S}^2$ moment directional memory, low-order spherical harmonic memory, candidate-set interaction ablations, edge-conditioned spatial GNNs, distributional Q_H heads, and privileged-teacher distillation. Dense GT candidate renders may be used only as privileged training signals in later ablations, never as V1 actor input. Point backbones, sparse convolutions, and continuous control are useful later ablations, not the first success criterion.

4 Learning Objective and Evaluation

The planned value model remains finite-candidate and masked. The candidate token, candidate value, and selected action are

$$\mathbf{u}_{t,i} = (\text{Transformer}_\theta(\mathbf{X}_t))_i \quad (29)$$

$$Q_{H,\theta}(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{q}_{t,i}) = \mathbf{w}_Q^\top \mathbf{u}_{t,i} \quad (30)$$

$$a_t^\theta = \arg \max_{i: m_{t,i}=1} Q_{H,\theta}(\mathbf{s}_t^{\text{cf0}}, \mathbf{z}_e, \mathbf{q}_{t,i}) \quad (31)$$

The first backup is fitted masked Double-Q [HGS15, Mni+13]. Here $d_t = 1$ at horizon termination, budget termination, or when no valid successor action exists:

$$i^* = \arg \max_{i: m_{t+1,i}=1} Q_{H,\theta}(\mathbf{s}_{t+1}^{\text{cf0}}, \mathbf{z}_e, \mathbf{q}_{t+1,i}) \quad (32)$$

$$y_t = r_t^e + \gamma(1 - d_t)Q_{H,\theta^-}(s_{t+1}^{\text{eff}}, z_e, \mathbf{q}_{t+1,i^*}) \quad (33)$$

$$\mathcal{L}_Q(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r,s') \in \mathcal{D}} m_{t,a}(Q_{H,\theta}(s_t^{\text{eff}}, z_e, \mathbf{q}_{t,a}) - y_t)^2 \quad (34)$$

All selected actions are re-evaluated by the oracle under identical roots, candidate budgets, and acquisition budgets. The minimum final evidence report must contain symbolic thresholds that are locked with the advisor before final experiments:

$$(S_{\min}, T_{\min}, N_{\min}) = (S_{\text{scenes}}, T_{\text{targets}}, N_{\text{roots}}). \quad (35)$$

Here the three symbols denote the minimum number of scenes, matched targets, and rollout roots. Coverage is reported against the full scale bar of 100 GT-mesh ASE scenes and 4,608 snippet windows, or against an explicit scene-level held-out subset if scale is blocked.

Surface	Required evidence
One-step scorer	rank correlation, top- k oracle hit, calibration, selected-candidate oracle RRI, target visibility, invalid fraction, grouped failure modes
Candidate and replay	strategy provenance, path increments, scene/target support fields, validity masks, reason codes, policy metadata, seed metadata, shuffled-candidate evaluation

Table 2: Compact evidence contract: quality is reported both as ranking performance and as replay/candidate integrity.

Policy	Actor input	GT decision	H	Role
π_{rand}	yes	no	1	lower reference over valid candidates
$\pi_{\text{learned-1}}$	yes	no	1	myopic learned target scorer
$\pi_{\text{oracle-1}}$	no	yes	1	one-step oracle upper bound
$\pi_{\text{oracle-look}}$	no	yes	H	non-myopic headroom estimate
π_Q	yes	no	H	learned recovery after oracle re-scoring

Table 3: Leakage-aware policy comparison. Report $J_e^{(H)}$, $G_t^{(H)}$, scene RRI, cost, invalidity, runtime, and coverage for each row.

5 What We Adopt From Prior Work

Source family	Role	Adopt	Do not adopt as core
VIN-NBV [Fra+25]	Closest quality-driven finite-candidate precedent.	Oracle point-mesh RRI labels, ordinal one-step ranking, learned myopic control.	Treating one-step greedy as enough for multi-step claims.
Project Aria / ASE / EFM3D [Eng+23, Met25, Str+24]	Logged egocentric sensing and mesh-supervised oracle substrate.	Poses, calibration, semidense points, frozen EVL/EFM evidence, OBB predictions.	GT geometry, GT OBBs, or dense labels as V1 actor input.
Greedy sensing [GK11, KSG08]	Reason greedy may be strong under diminishing returns.	Measure oracle-lookahead headroom before claiming planning gain.	Assume non-myopic learning is useful without a headroom test.
CORAL and set models [CMR19, Lee+19]	Ranking and permutation-aware candidate scoring.	Ordinal one-step target scorer; candidate-token Q_H with masked set output.	Unbounded architecture search before target/RL evidence is stable.
Offline value learning [FMP19, HGS15, KNL21, Kum+20, Mni+13]	Replay, overestimation control, and offline support constraints.	Masked fitted Double-Q first; IQL/CQL/BCQ only as later ablations.	Optimizing invalid or unsupported actions.
GenNBV, Hestia, 3DGS, SceneScript [Ave+24, Che+24, JLD24, Lu+26]	Continuous, hierarchical, uncertainty, semantic, and global-memory contrast.	Design pressure for post- Q_H bridge work and discussion.	Replacing target point-mesh RRI with coverage, uncertainty, or semantic proxies as the main result.

Table 4: Literature role ledger: each source family justifies a design choice or a deferred bridge; it does not broaden the thesis claim.

6 Roadmap, Risks, and Decisions

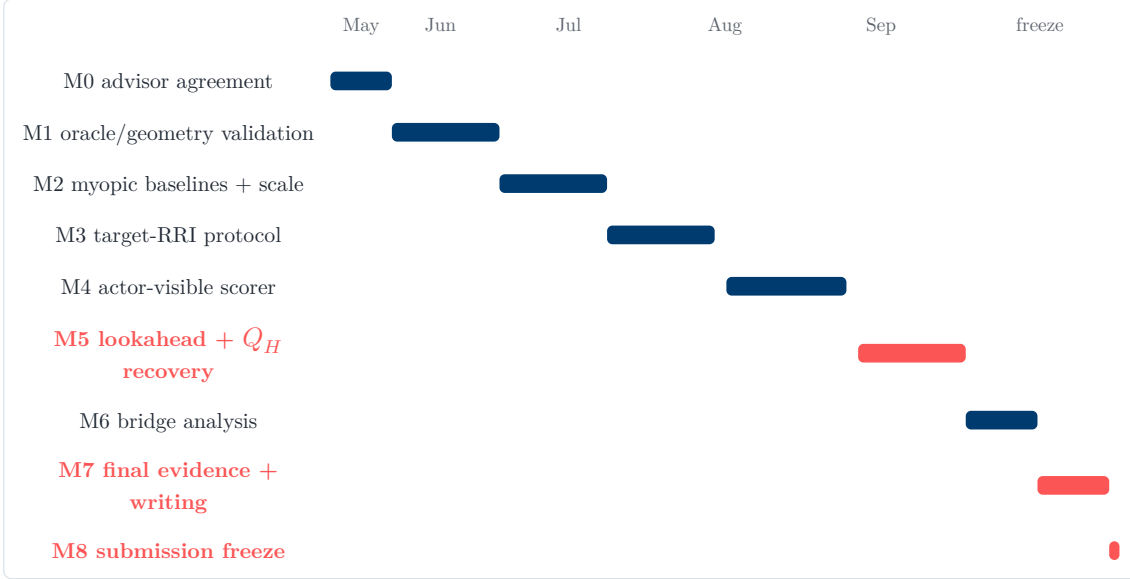


Figure 1: Typst-native thesis Gantt, 2026-04-29 to 2026-09-30. Red bars mark interpretation-critical evidence and submission freeze.

Three risks decide interpretation. If M1 geometry or oracle labels fail, the thesis becomes a contract and one-step-scoring study. If target matching is sparse or ambiguous, target RRI is reported only on validated subsets with unmatched counts. If Δ_{look} is near zero, the thesis reports an effectively myopic candidate/objective regime rather than promoting continuous RL as a substitute.

Open advisor decisions are deliberately narrow: final scene-level split, numeric minimum scale $(S_{\min}, T_{\min}, N_{\min})$, pass/fail threshold for η_Q , exact target-match thresholds $(\tau_{\mu}, \tau_{\text{gap}})$, and the first actor-visible crop descriptor ablation. These choices change the evidence bar, not the thesis spine.

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